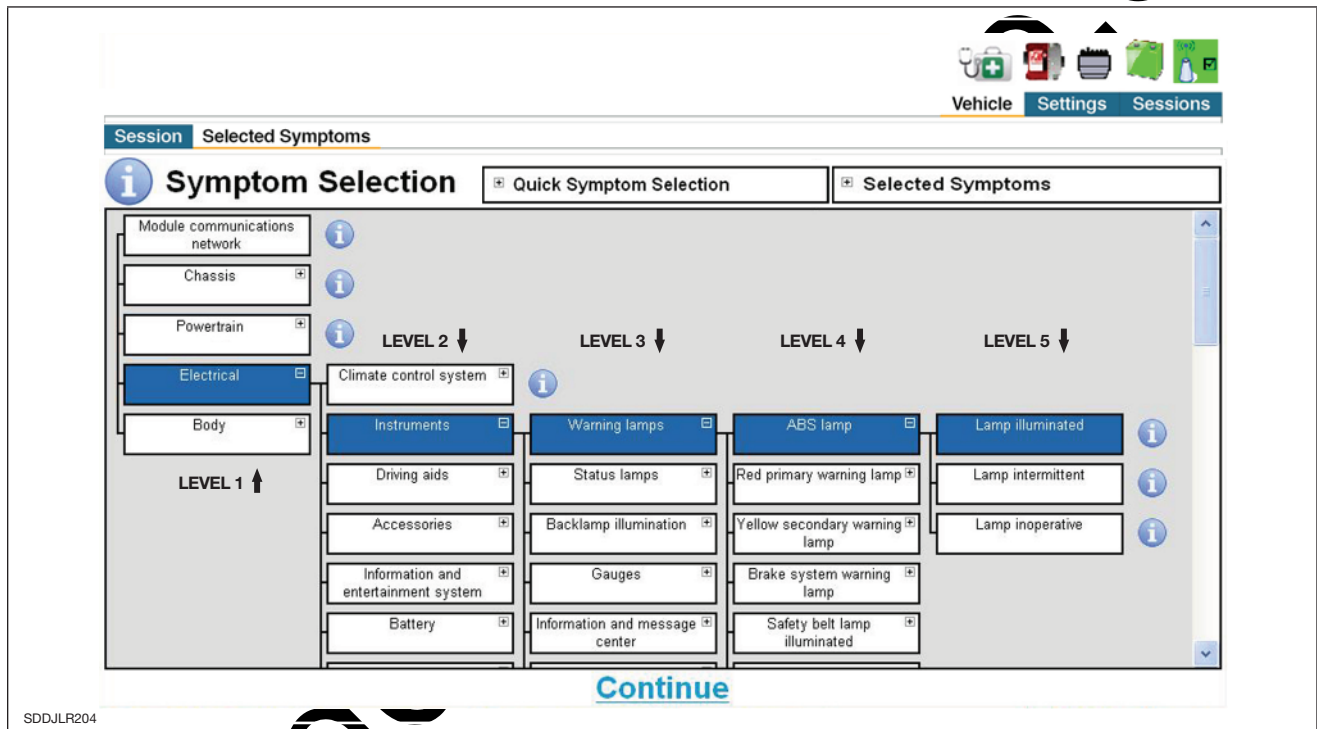


Symptom Selection

This screen displays the currently released Symptom Maps for the selected vehicle.

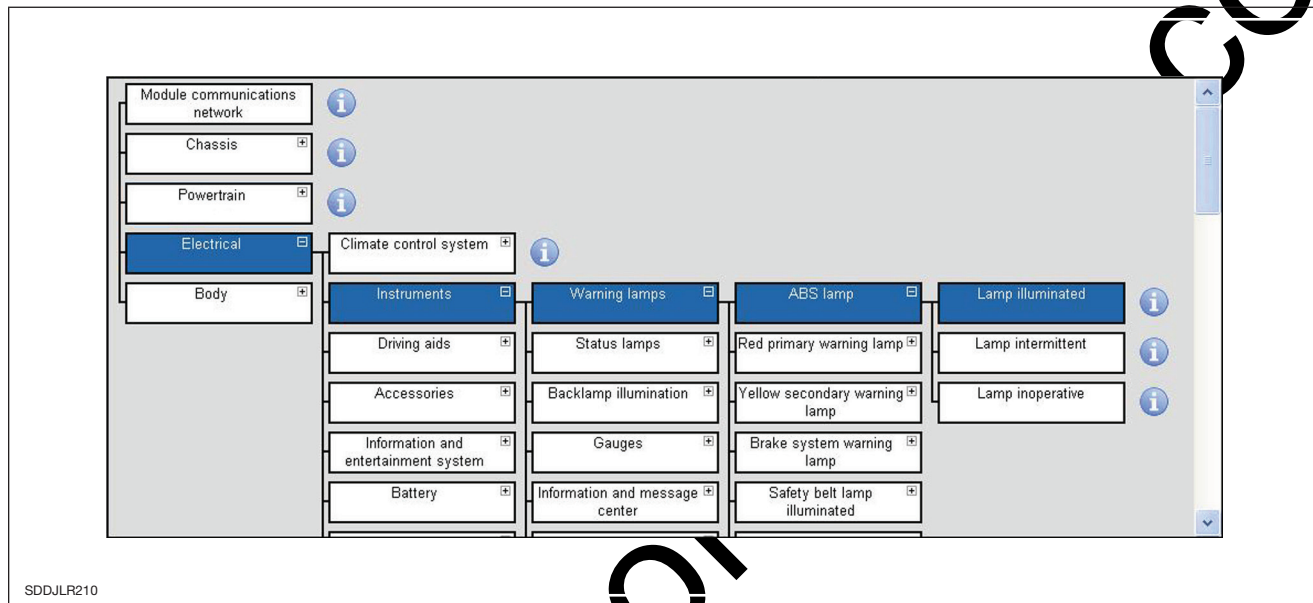


NOTE: The user must make selections down to at least Level 3 before being able to continue with diagnosis. Additional levels may be available, to allow the operator to further isolate the area of concern. It is advisable the operator isolates a symptom as far as possible, as the remedial recommendations will be more specific.

The first three Levels identify the Systems/Areas of the vehicle. A route should be selected through these first three Levels, according to the information that is available and relevant to the concern under investigational. If only limited information or no specific symptom is available, then diagnostics should be started at this point.

Levels 4 and 5 identify specific symptoms for the selected Systems/Areas of the vehicle. These should be used only when relevant to the identified concern, to allow the SDD to determine the most appropriate diagnostic strategy and offer the most appropriate remedial actions.

This screen also contains 'Quick Symptom Selection' and 'Selected Symptoms' drop-down menus.



In the example shown:

If the information available to the operator is that various warning lamps have been illuminated on the Instrument Cluster, the appropriate selection will be:

ELECTRICAL > INSTRUMENTS > WARNING LAMPS.

If diagnostics are then started from this point, data will be displayed for all warning lamps.

If the information available to the operator is more specific and identifies that only the ABS warning lamp is illuminated, the appropriate selection will be:

ELECTRICAL > INSTRUMENTS > WARNING LAMPS > ABS LAMP > ABS LAMP ILLUMINATED.

When diagnostics are started from this point, data will be displayed for the ABS warning lamp only.

When navigating through the Symptom Maps, a single click will select the required option and display any further options available; a double click will deselect and close the further options.

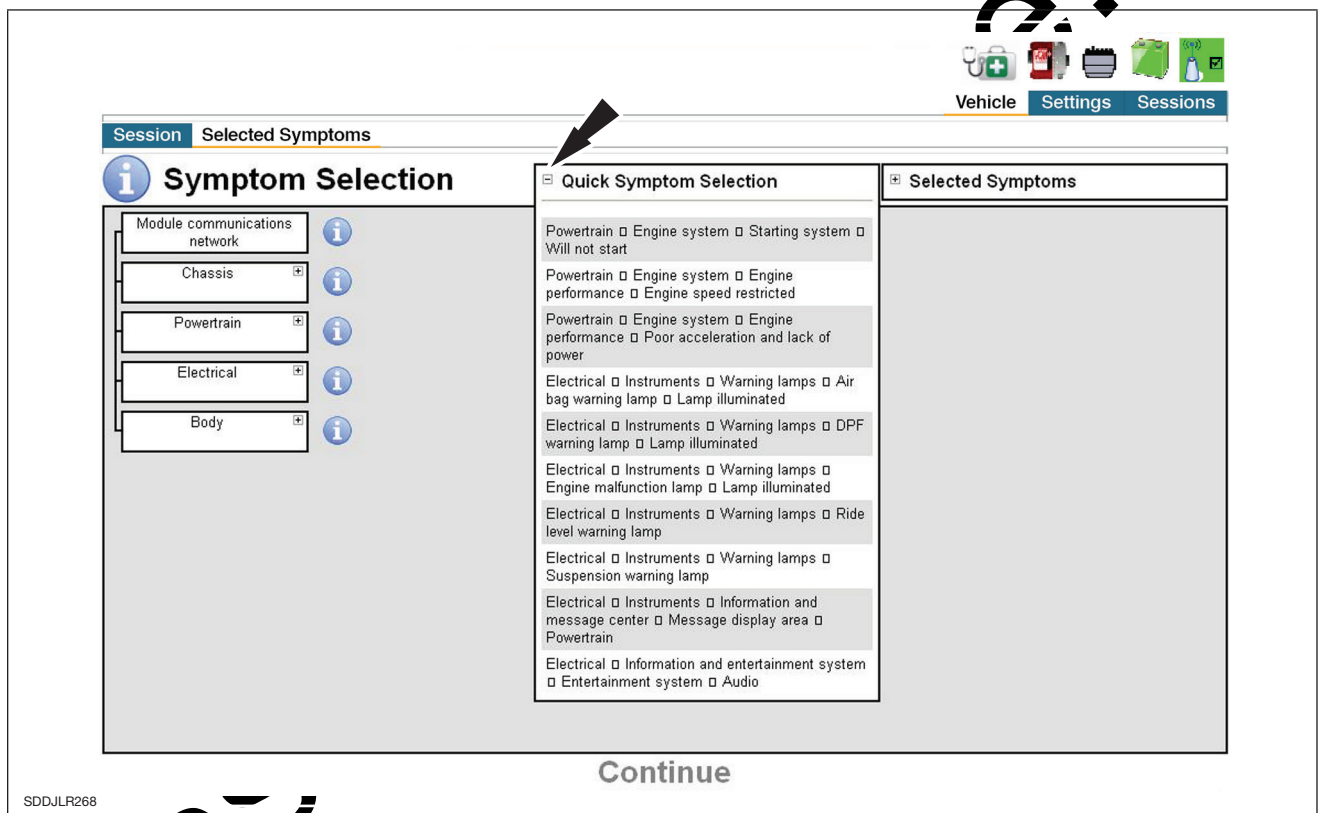
Multiple Symptoms can be selected for one session, and therefore SDD will carry out data analysis based on all symptoms selected.

Once all relevant symptoms have been selected, click 'Continue' to display the Related Events screen.

NOTE: This example is used to illustrate the fact that, if the MIL is illuminated, the user must list this as a symptom. Failure to select symptoms accurately will reduce the effectiveness of SDD.

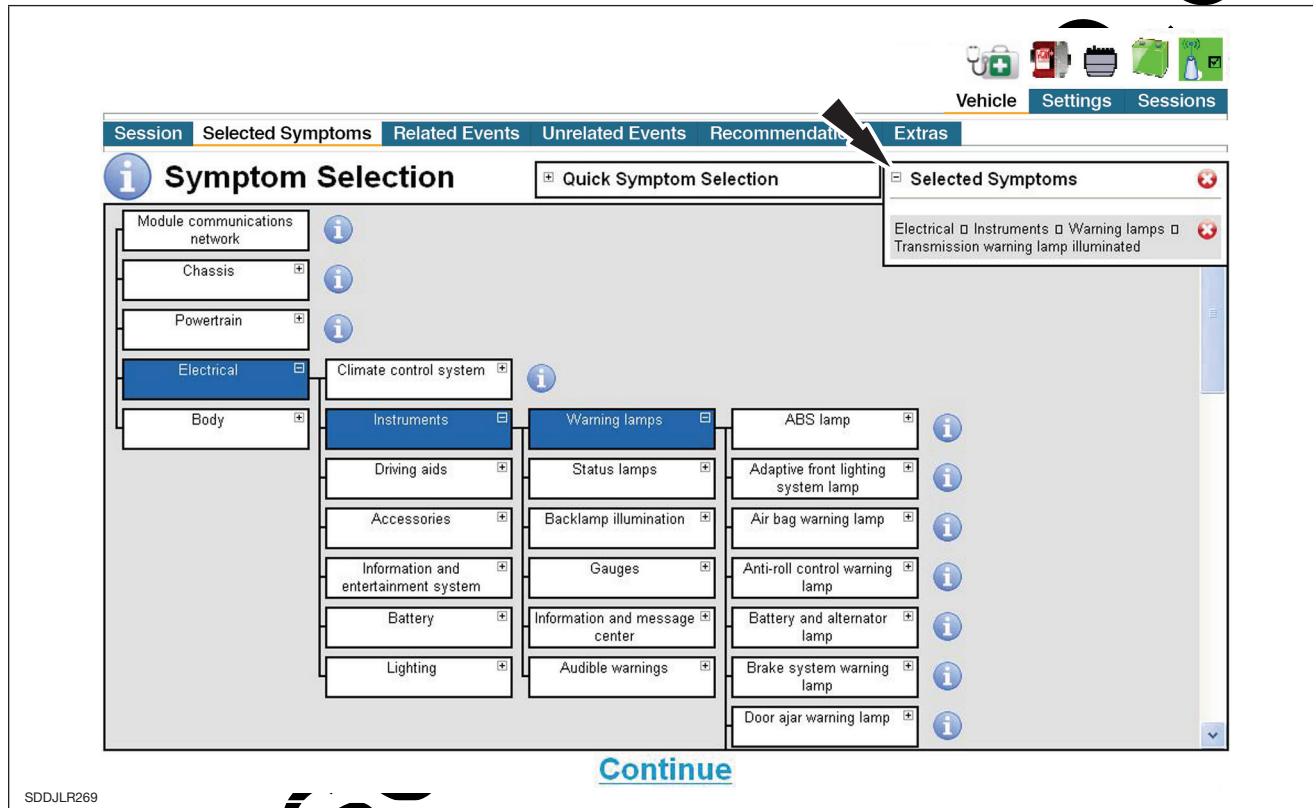
Quick Symptom Selection

The 'Quick Symptom Selection' drop-down menu provides an alternate way to select a specific symptom without going through the entire symptom map.



Selected Symptoms

The 'Selected Symptoms' drop-down menu displays the symptoms that have been selected as described previously.



Improving SDD Effectiveness when Selecting Vehicle Symptoms

SDD is designed to link DTCs and provide recommendations based on *Customer Facing* and/or the *most significant* symptoms. Do not try to short-cut or refine the customer's reported symptoms based on expected failures. If the primary or most obvious Customer Facing symptom is an MIL or message center warning, those symptoms **MUST** be included in your symptom selection inputs using SDD.

While it might seem logical when selecting 'Air Suspension is inoperative' that there is really no need to select the symptom 'Message center warning', it is important to do so. Otherwise SDD may exclude DTCs or recommendations that relate to failures known to trigger a warning message or MIL.

Be sure to select any additional symptoms observed during a road test in addition to the most obvious symptoms and customer reported info.

Altering the selected symptoms adjusts the filtering of the DTCs. Selecting fewer symptoms makes the filter 'coarser' while selecting more symptoms makes the filter 'finer'.

- If very few or no DTCs and recommendations are listed, try restricting your symptom selections to Level 3 in the symptom tree.
- If too many DTCs or recommendations are listed, try selecting symptoms to Level 4 or 5, using care not to select an inaccurate symptom.
- To be sure that you are only working with your intended symptom list: open the 'Selected Symptoms' drop down box at the upper right portion of the Symptom Selection tab to review all currently selected symptoms.

Avoiding Common SDD Symptom Selection Errors

Powertrain > Engine system > Starting system > Start-stop system:

Note that the Start-Stop System is a fuel conserving technology used on diesel vehicles sold in other markets. This is not associated with the 'Start Button' used to actuate the ignition on Smart Key equipped vehicles.

Warning Lamps, Status Lamps, and Message Center

Use care when entering symptoms related to customer complaints of MIL lights and warnings. When a light and a message are reported or observed, both of these symptoms must be included.

When viewing the symptom categories within the Electrical > Instrument Cluster section, there are 3 classes of 'driver information' symptoms:

- Warning Lamps (red)
- Warning Messages (in Message Center)
- Status Lamps (yellow)

While it is easy to distinguish between a Warning Message and a Warning Lamp that is currently active, if the problem is intermittent, the service writer must determine through careful questioning of the customer if a warning lamp, warning message, or both were displayed.

Also, not every MIL is considered a Warning Lamp since some indicators on the instrument cluster are used to indicate the status of a system. Examples of Status Lamps include the Passenger Seat Occupancy Status Lamp and the TPMS Status Lamp.

Using the Passenger Seat Occupancy Status Lamp as an example: this light is only used to indicate the status of the passenger seat. When lit, it simply indicates that the SRS system has determined that based on the current inputs, the passenger restraints will not be deployed. It does not mean the system has detected a failure that requires a driver warning. For this reason, the lamp is classified as a Status Lamp.

Similarly, the amber TPMS indicator is used to inform the driver that the tire pressure requires attention. While this light may remain lit when a system malfunction exists, it is considered a Status Lamp since its primary function is to inform the driver of the tire pressure status.

Recommendation:

When selecting SDD symptoms for lamps that are primarily considered a Status Lamp, but are also known to have a secondary function as a Warning Lamp, be sure to select the symptoms under both categories in the symptom selection map.

Related Events

Clicking on the 'Related Events' tab will launch the Summary Event View, which displays all the events relevant to the symptoms the user has entered. The default screen is sorted by distance, but results can be sorted by any of the parameters by clicking on the column heading. A small red triangle indicates which parameter the results are sorted by.

Related Events

SDD has detected 5 DTC events which are related to the customer symptoms you have selected. These are detailed below and should be investigated first.

Event	Distance	Time	DTC count	Relevance
15	Events at 14,415miles	Events at 117 days	3	34%
14	Events at 14,138miles	Events at 116 days	6	2%
13	Events at 13,484miles	Events at 114 days	1	4%
12	Events at 13,055miles	Events at 112 days	3	4%
10	Events at 12,101miles	Events at 92 days	3	65%

These related events can be viewed by selecting the regions of the timeline on the next screen.

All other DTC events are accessible via the "Unrelated Events" tab. These are less likely to explain the customer symptoms you have entered.

To view the related events, click any event number listed above, or select continue.

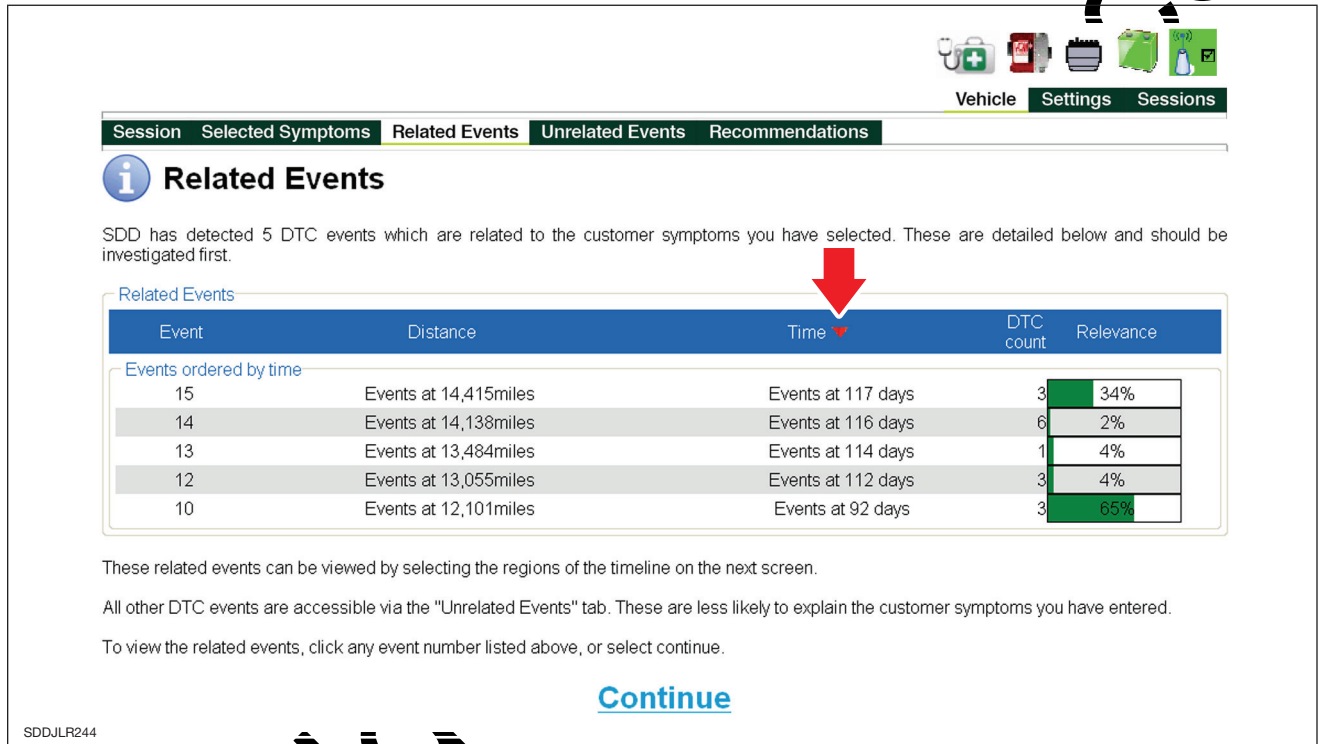
EVENT NUMBERS [Continue](#)

SDDJLR243

Clicking on an event number will load the Related Events Timeline view.

NOTE: DTC count refers to the number of DTC events at that mileage event.

Results Sorted by Time



Vehicle Settings Sessions

Session Selected Symptoms **Related Events** Unrelated Events Recommendations

Related Events

SDD has detected 5 DTC events which are related to the customer symptoms you have selected. These are detailed below and should be investigated first.

Event	Distance	Time	DTC count	Relevance
Events ordered by time				
15	Events at 14,415miles	Events at 117 days	3	34%
14	Events at 14,138miles	Events at 116 days	6	2%
13	Events at 13,484miles	Events at 114 days	1	4%
12	Events at 13,055miles	Events at 112 days	3	4%
10	Events at 12,101miles	Events at 92 days	3	65%

These related events can be viewed by selecting the regions of the timeline on the next screen.

All other DTC events are accessible via the "Unrelated Events" tab. These are less likely to explain the customer symptoms you have entered.

To view the related events, click any event number listed above, or select continue.

[Continue](#)

SDDJLR244

The time value in days is intended to be the life of the vehicle (although it tends to get reset unintentionally on some vehicles such as the LM Range Rovers).

The benefit of the time value is in comparing DTCs that were set at the same mileage, but on different days. Consider the following example:

The vehicle breaks down on the road, sets DTCs, gets towed home on a Friday evening, and sits over the weekend. The technician gets to the vehicle on the following Tuesday, finding that the battery was discharged because flashers were left on.

The technician finds that numerous DTCs have been set, but 2 are from Friday (Day 10), the rest are from 4 days later on Tuesday (Day 14) and have probably been caused by the dead battery.

To find out what is actually wrong with the vehicle, the technician would focus on the 2 DTCs from Day 10.

Results Sorted by Relevance

This view sorts results by their relevance to the symptoms entered.

Related Events

SDD has detected 5 DTC events which are related to the customer symptoms you have selected. These are detailed below and should be investigated first.

Event	Distance	Time	DTC count	Relevance
10	Events at 12,101miles	Events at 92 days	3	65%
15	Events at 14,415miles	Events at 117 days	3	34%
13	Events at 13,484miles	Events at 114 days	1	4%
12	Events at 13,055miles	Events at 112 days	3	4%
14	Events at 14,138miles	Events at 116 days	6	2%

These related events can be viewed by selecting the regions of the timeline on the next screen.

All other DTC events are accessible via the "Unrelated Events" tab. These are less likely to explain the customer symptoms you have entered.

To view the related events, click any event number listed above, or select continue.

[Continue](#)

SDDJLR245

Clicking on an event number will open the Related Events Timeline view, which allows the results to be viewed on a timeline with several options for sorting (indicated by the small red triangle).

Results Sorted by Distance

Related Events (Distance) [Summary Event View](#)

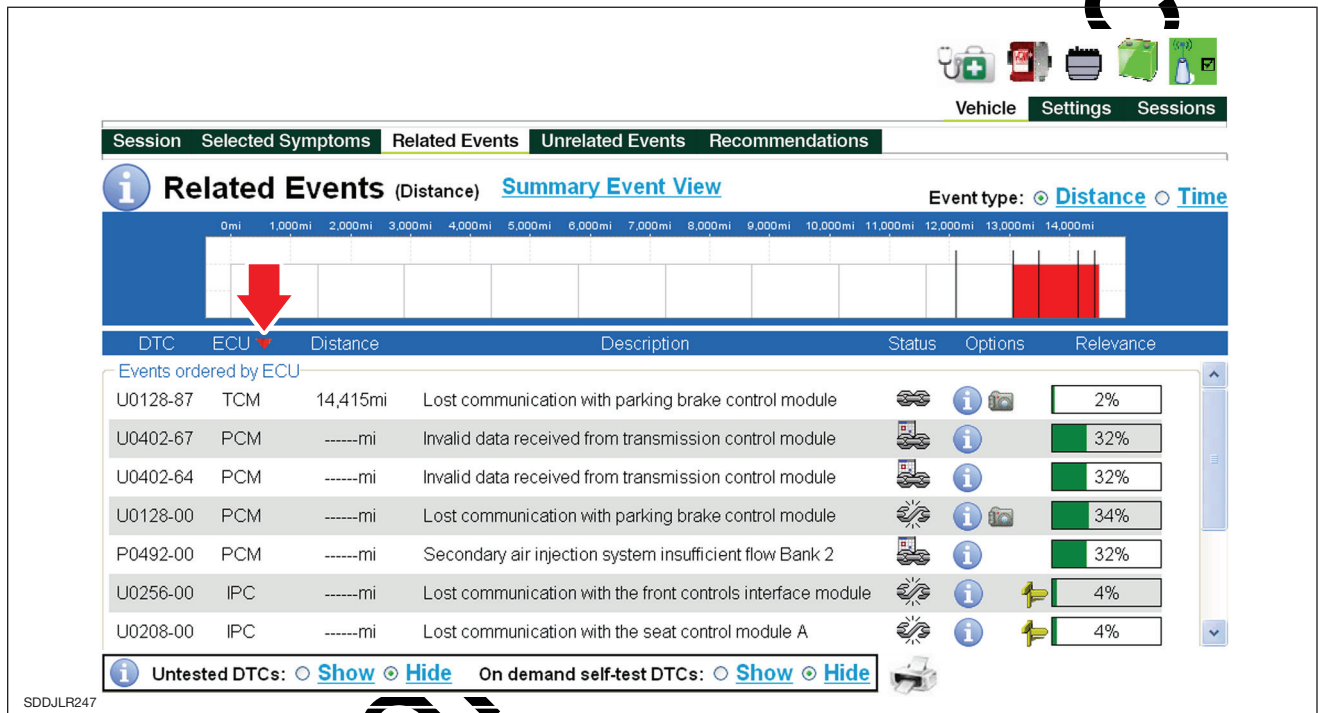
Event type: ☒ Distance ☐ Time

DTC	ECU	Distance	Description	Status	Options	Relevance
Events at 14,415mi						
U0128-00	IPC	14,415mi	Lost communication with parking brake control module			34%
U0128-87	TCM	14,415mi	Lost communication with parking brake control module			2%
U0128-00	ABS	14,415mi	Lost communication with parking brake control module			4%
Events at 14,138mi						
U3000-87	BPM	14,138mi	Control module			0%
U0184-00	ICM	14,138mi	Lost communication with audio unit			2%
U0186-00	ICM	14,138mi	Lost communication with the audio amplifier			2%

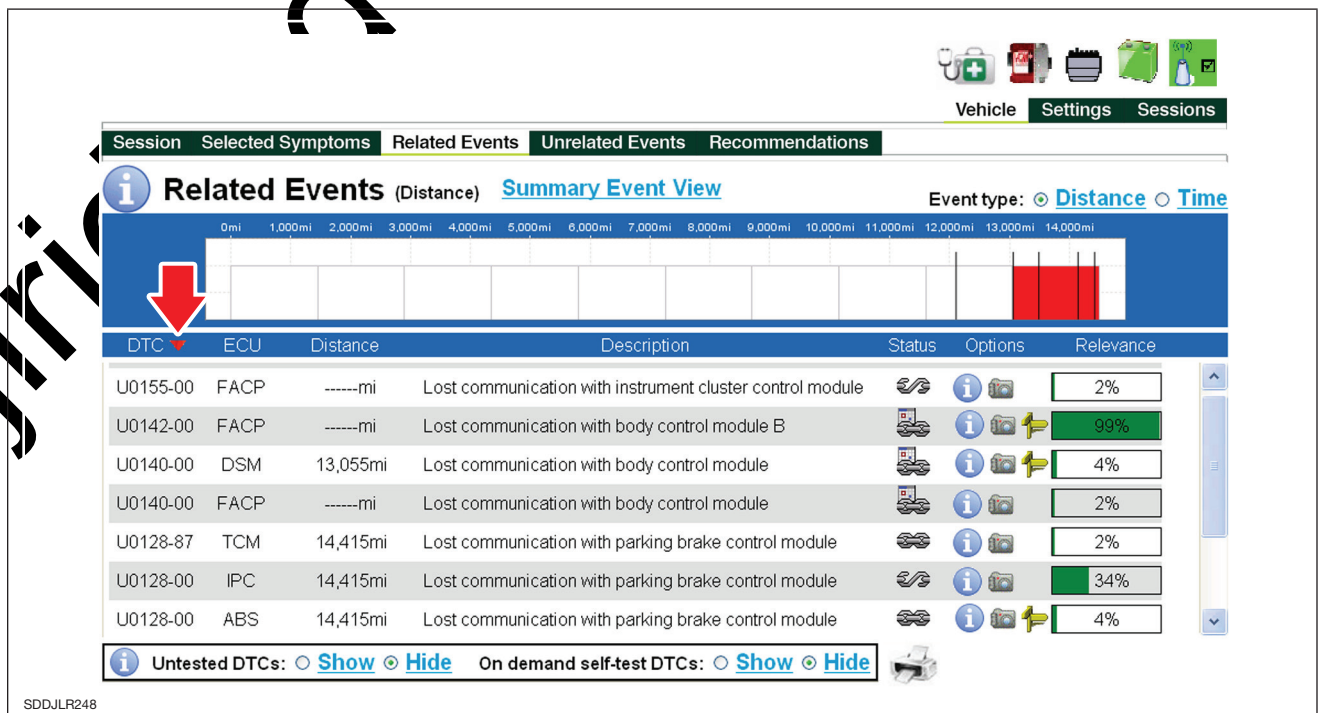
Untested DTCs: ☐ Show ☒ Hide **On demand self-test DTCs:** ☐ Show ☒ Hide

SDDJLR246

Results Sorted by Control Module (ECU)



Results Sorted by DTC



Related Events Timeline

EVENTS

SELECT EVENT TYPE (DISTANCE SELECTED)

Vehicle Settings Sessions

Session Selected Symptoms Related Events Unrelated Events Recommendations

Related Events (Distance)

Event type: Distance Time

Order events by: Distance Relevance

DTC	ECU	Distance	Description	Status	Options	Relevance
B1001-07	ECU	396mi	High temperature			
U0100-00	ECU	396mi	Module data received from ABS control module			
U0100-00	ECU	396mi	Filter apply: communication bus			
U0100-00	ECU	396mi	Lost communication with parking assist control module			
U0100-14	ECU	396mi	Control module output power A			
U0100-00	ECU	396mi	Lost communication with ABS control module			

Untested DTCs: Show Hide On demand self-test DTCs: Show Hide

SEGMENT SELECTED

SDDJLR206

The Related Events timeline displays 'Events' as a heavy black vertical line. An event is a point in time at which a group of DTCs were triggered simultaneously. Events can be viewed by Distance and Time by selecting either the 'Distance' or 'Time' button above the timeline.

Clicking on segments of the timeline will display the DTCs triggered by the events that occurred during that segment (Distance or Time, depending on which Event type is specified). When a segment is selected, it will be highlighted in red.

Only the events that contain one or more DTCs related to the currently selected symptom(s) will be displayed in the Related Events timeline.

The DTC listing in the Related Events screen includes the following for each DTC:

- Module that flagged the DTC
- Distance or Time at which it was set
- Brief description
- Status
- Options, which may include:
 - DTC Help information
 - Snapshot data
 - Guided Diagnostics (Recommendations)
- Relevance

By default, SDD displays only the most recent event segment in the timeline, as indicated by the red block. Each segment may contain more than one event. Users must select and deselect portions of the timeline to view each event individually or in small groups when clustered within a single segment. This allows technicians to quickly determine which DTCs are linked to a common event in the vehicle's history. The following example shows an earlier event selected.

Related Events (Distance) Event type: ☒ Distance ☐ Time

Order events by: ☒ Distance ☐ Relevance

DTC	ECU	Distance	Description	Status	Options	Relevance
Events at 56mi						
U0415-00	HCMB	57mi	Module data received from ABS control module			
B1A02-13	AAM	57mi	Speeder 2 circuit			
B1A01-13	AAM	57mi	Speeder 1 circuit			
U2013-24	DDM	57mi	Switchgear			
U0140-00	SCME	57mi	Lost communication with body control module			
B12BD-31	IPMB	57mi	No description available			
B1D07-15	DDM	57mi	Switchgear circuit			

Untested DTCs: ☐ Show ☐ Hide On demand self-test DTCs: ☐ Show ☐ Hide

SDDJLR207

The user can select two segments at one time.

Related Events (Distance) Event type: ☒ Distance ☐ Time

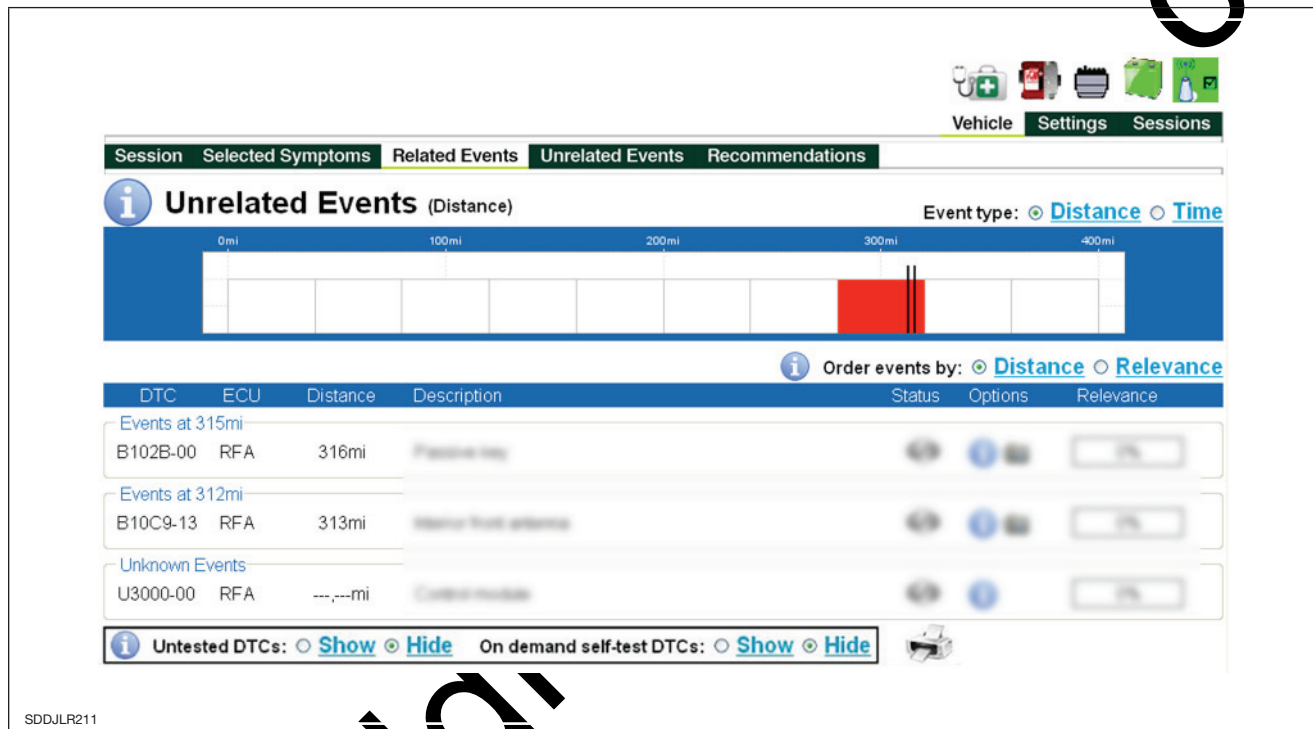
Order events by: ☒ Distance ☐ Relevance

DTC	ECU	Distance	Description	Status	Options	Relevance
Events at 396mi						
B12BE-31	IPMB	396mi	No description available			
C111B-15	BCM	396mi	Left door open			
B1247-15	BCM	396mi	Power window signal			
B130E-15	BCM	396mi	No description available			
B1B01-87	BCM	396mi	Key transponder			
U0415-00	RCM	396mi	Module data received from ABS control module			
U1D002-86	BCM	396mi	Left door open			

Untested DTCs: ☐ Show ☐ Hide On demand self-test DTCs: ☐ Show ☐ Hide

SDDJLR208

The user should also view Unrelated Events to learn about other DTCs in the vehicle's history.



Advantages of Viewing DTCs on a Timeline

DTCs are associated with Distance or Time-based 'Events', as displayed in the timeline when viewing the Related or Unrelated Events Screens.

While SDD has the ability to link DTCs to symptoms, the technician must evaluate how the timing of the events relates to the customer complaint. This feature will be most effective when service advisors accurately document the detail of the customer complaint on the Repair Order, especially in association with warning lamps and electrical failures.

For example, if the Repair Order states 'MIL illuminated at about 35,000 miles', or 'Audio began cutting out intermittently shortly after the 15K service', the technician can use the Event timeline sorted by 'Distance' to effectively associate one group of DTCs with a specific customer complaint.

Because Events can also be sorted by Time, the following examples would be equally useful to the technician:

- MIL illuminated 2 weeks ago
- Audio has been cutting out intermittently for about a month now.

The option to sort Events by time provides a significant advantage for technicians when multiple DTCs have been stored at the same mileage. Since the time sorting is possible down to one-tenth of a second, it is most effective for identifying the root cause of multiple failures. This becomes most significant when it involves network failures within the vehicle where a few 'hard faults' cascade out to multiple modules as network info U04XX DTCs.

The technician can sort the Events by time and look at the first 3 – 5 DTCs in the list as the most significant issues and likely root cause.

‘Two-Trip’ DTCs

When a DTC is set in a vehicle system that supports the ISO14229 diagnostic standard, information about time/mileage is processed from CAN data and recorded with the ‘global snapshot’ record for that DTC. When SDD processes DTCs from the vehicle, the information in the global snapshot is read and used to plot each DTC on the timeline.

Note the following details regarding this global snapshot data for mileage/time.

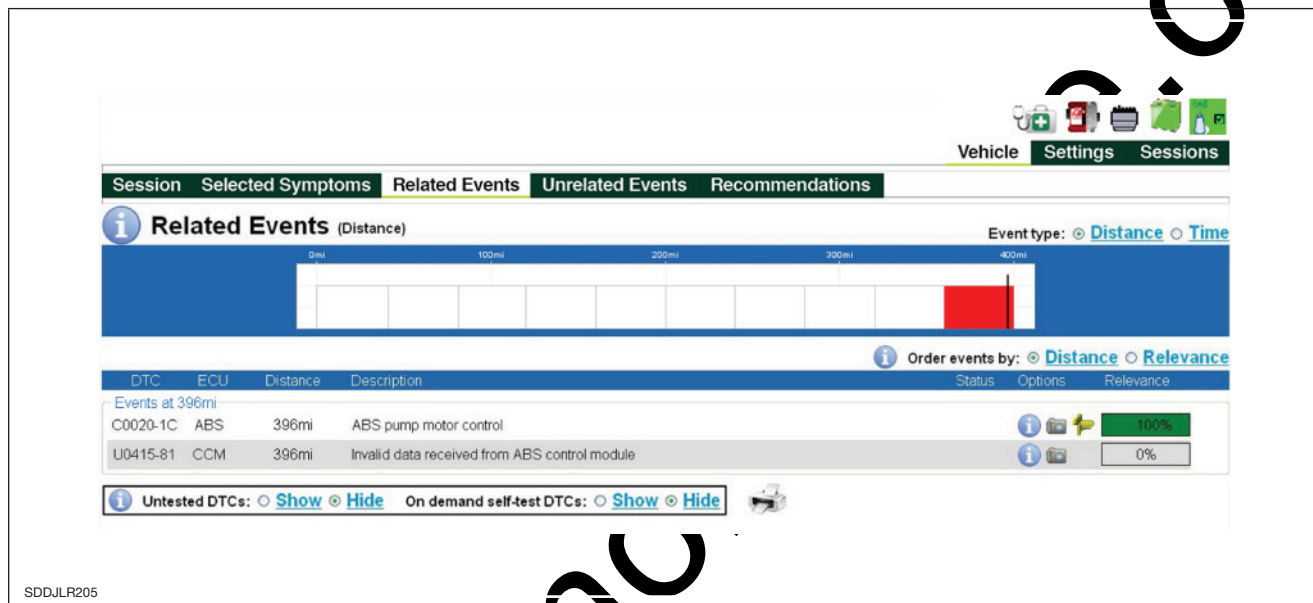
1. Since the global snapshot data is only recorded when the DTC is first set, any single DTC can only be listed once on the timeline regardless of how many more times the DTC has occurred.
2. Since the global snapshot data is derived from information broadcast on the CAN bus, there can be times when conditions prevent the recording of mileage/time when a DTC is set.

If a DTC is being recorded when the vehicle CAN bus is being initialized, or is not communicating, the DTCs global snapshot record may not contain time and mileage data to allow SDD to project it on a timeline. Such DTCs will be listed as an ‘Unknown Event’.

3. When investigating customer concerns about Check Engine MILs, consider the impact of PCM/OBDII ‘Two-Trip’ DTCs, such as those for Evaporative Emissions Leaks.
 - Since each DTC will only record global snapshot data from the first time it was detected, Two-Trip DTCs that do not trigger a Check Engine MIL on the first occurrence may not appear on the timeline at the point where the customer has reported that the Check Engine MIL first illuminated.
 - For example, if the PCM first detects a P0456 at 5500 miles, this code will be set with 5500 miles as the global snapshot data. If the problem is intermittent and/or the diagnostic monitoring conditions prevents this code from being detected, or set a second time in close proximity to the original failure at 5500 miles, the Check Engine MIL may not be illuminated for several hundred or even thousands of miles, depending on the customer’s driving habits.
 - When the vehicle condition finally degrades to the point where the ‘Two-Trip’ requirements are met to trigger the Check Engine MIL, the customer may report that the failure occurred at 7500 miles (for example), but the strategy in the vehicle to record the global snapshot record at the time of the first occurrence will place this DTC further back in the timeline at the 5500 mile mark.
 - Refer to the OBDII systems documentation on TOPIx to determine which PCM DTCs are regarded as ‘Two-Trip’ failures to trigger a MIL.

With the exception of these Two-Trip DTCs in the PCM, in most cases it is reliable to associate the customer’s story with the events in the timeline for all other areas of vehicle operation, since most other systems and DTCs will instantly trigger an MIL or message in the instrument cluster.

ABS DTC C0020-1C Example



In the example shown:

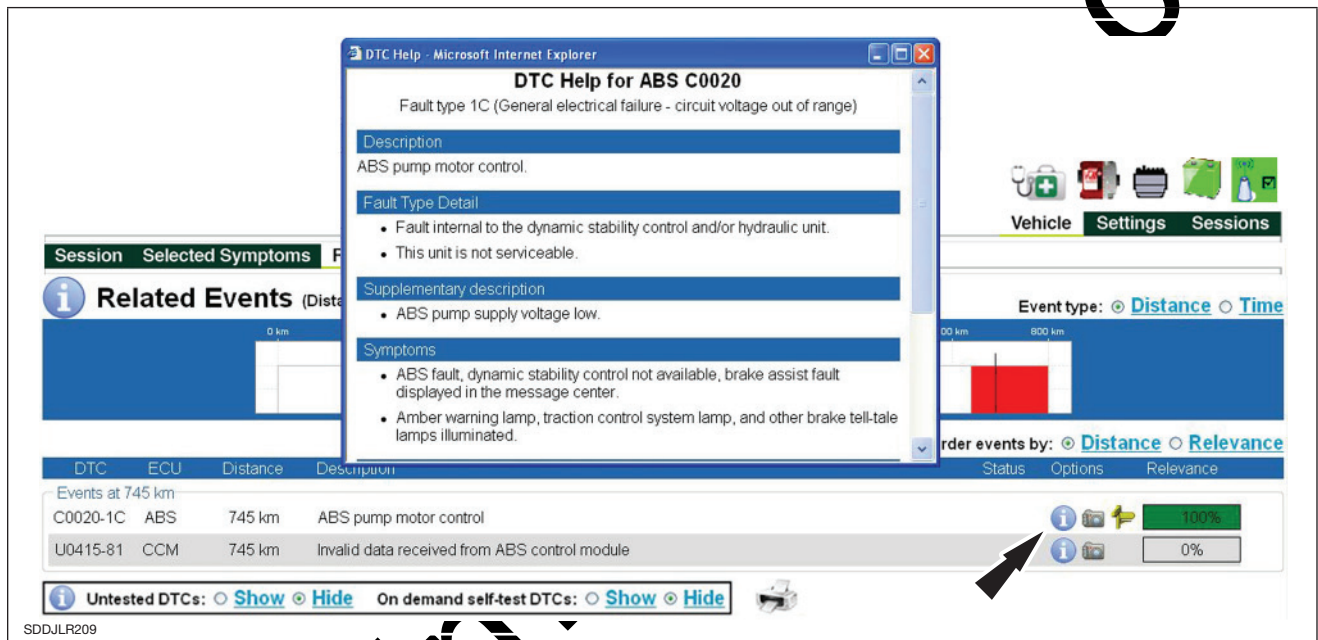
- DTC C0020-1C 'ABS pump motor control' was flagged by the ABS module at 396 miles.
 - DTC Help information (i icon), Snapshot data (camera icon) and Guided Diagnostics (signpost icon) are available
 - This DTC is 100% Relevant to the Selected Symptoms

This DTC's 'Relevance' rating indicates that there is a 100% probability that it is the fault that caused the Selected Symptoms. 'Guided Diagnostics' indicates that there is a Pinpoint Test available, which is highly recommended. Clicking on the Guided Diagnostics icon will also display a page with a filtered list of recommended tools related to this DTC specifically.

NOTE: With SDD, Pinpoint Tests are no longer launched from the DTC displays but instead will appear as Recommended Candidates once the 'recommendations' tab is selected.

NOTE: The Pinpoint Tests are specific to the Selected Symptoms rather than an individual DTC.

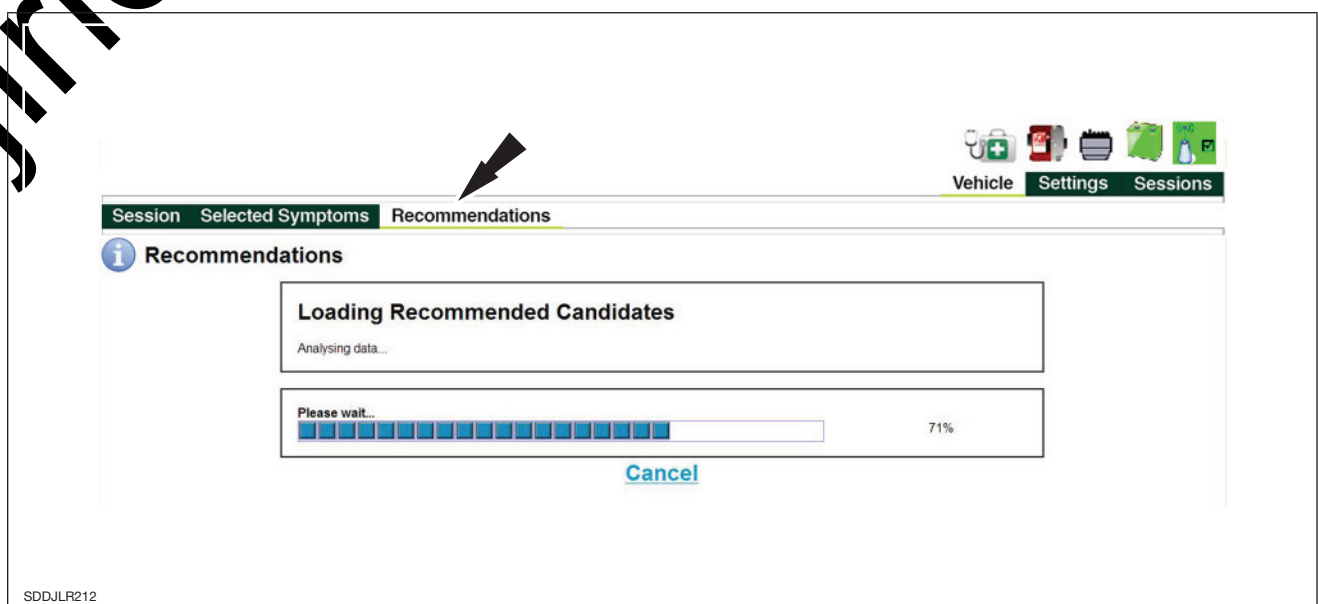
Clicking on the Information icon (i) will display the DTC Help pop-up.



The pop-up will show the following information:

- DTC number
- Fault type
- Description
- Fault Type detail
- Supplementary description
- Symptoms

Click on the 'Recommendations' tab to load Recommended Candidates.



Candidates are listed in order of priority.

Recommendations (Ordered by priority)

Highly Recommended

Type	Description	Status	Options
	ABS PUMP MOTOR POWERFEED TEST	Not run	Run
	TSB XK419-02 - Falsely Flagged Diagnostic Trouble Codes (DTCs)	Not read	Read
	TSB XK100-005 - Wheel Speed Sensor Concerns Diagnosis	Not read	Read
	TSB XK100-006 - CAN Faults Relating To The ABS Module Diagnosis	Not read	Read
	TSB XK100-007 - Internal Control Module Faults Relating To The ABS Module Diagnosis	Not read	Read
	TSB JTB00017 - Additional Diagnostic For Instrument Cluster	Not read	Read

Recommended

Type	Description	Status	Options
	SSM [summary description] - Approved Accessories	Not read	Read
	Powertrain - Purge valve self test	Not run	Run

SDDJLR213

The highest priority candidate for the Selected Symptom is the ABS PUMP MOTOR POWERFEED TEST. There are also several Highly Recommended Technical Service Bulletins.

Technical Service Bulletins may include a relevant diagnostic process.

JAGUAR

MODEL: 2007 XK DATE: 07 Dec 2006 NUMBER: XK100-003

SERVICE TECHNICAL BULLETIN

SECTION: 100-00 - GENERAL INFORMATION

CAN Faults Relating to the ABS Module Diagnosis

AFFECTED VEHICLE RANGE:

New XK VIN: B00001 onwards
Model Year: 2007 onwards

CONDITION SUMMARY:

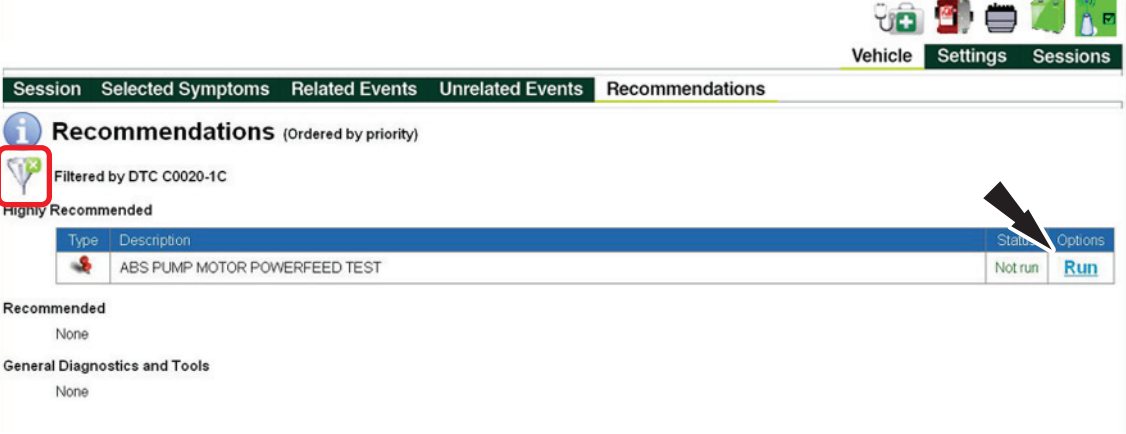
DIAGNOSTIC GUIDANCE FOR CAN FAULTS RELATING TO ABS MODULE DIAGNOSIS

[Close](#)

SDDJLR214

Running the Pinpoint Test

To run the ABS PUMP MOTOR POWERFEED TEST, click on 'Run' in the right-hand 'Options' column.



The screenshot shows the IDS-SDD-JLR software interface. At the top, there are tabs for 'Session', 'Selected Symptoms', 'Related Events', 'Unrelated Events', and 'Recommendations'. The 'Recommendations' tab is active. Below the tabs, there is a section titled 'Recommendations (Ordered by priority)'. A red box highlights a funnel icon in the top left of this section. To the right of the funnel icon, it says 'Filtered by DTC C0020-1C'. Below this, there is a table with the following data:

Type	Description	Status	Options
	ABS PUMP MOTOR POWERFEED TEST	Not run	Run

Below the table, there are sections for 'Highly Recommended', 'Recommended', and 'General Diagnostics and Tools', all of which show 'None'.

SDDJLR215

Follow the on-screen instructions in the successive windows to complete the test.

NOTE: Clicking on the 'funnel' icon will remove filtering and display all recommendations.

Vehicle Settings Sessions

Session Selected Symptoms Related Events Unrelated Events Recommendations Extras

ABS PUMP MOTOR POWERFEED TEST

Pinpoint | Reference | Flow chart | Options

Note:
This pinpoint test is intended to diagnose the following:

- Open circuit in the Pump supply circuit
- Open circuit in the Pump - Ground circuit
- Short circuit to ground in the Pump supply circuit
- Faulty ABS control module.

Note:

- Check vehicle battery condition

Continue...

SDDJLR216

Vehicle Settings Sessions

Session Selected Symptoms Related Events Unrelated Events Recommendations Extras

ABS PUMP MOTOR POWERFEED TEST

Pinpoint | Reference | Flow chart | Options

PWR31-1501 : PUMP SUPPLY FUSE CHECK

- Remove Fuse 20 from front power distribution box.
Check if Fuse 20 (front power distribution box) is blown.
- Is the (front power distribution box) - Fuse 20 undamaged?

Yes	No
Go to PWR31-1502	Go to PWR31-1506

SDDJLR217

The screenshot shows a diagnostic interface for the 'ABS PUMP MOTOR POWERFEED TEST'. At the top, there are icons for Vehicle, Settings, and Sessions. Below these are tabs for Session, Selected Symptoms, Related Events, Unrelated Events, Recommendations, and Extras. The main heading is 'ABS PUMP MOTOR POWERFEED TEST'. Underneath, there are links for Pinpoint, Reference, Flow chart, and Options. The test title is 'PWR31-1502 : LATCHED CONNECTOR CHECK.'. The instructions are: 'Check the following connectors are correctly latched. --ABS control module --FPDB'. The question is 'Are the connectors correctly latched?'. The response options are 'Yes' (Go to PWR31-1503) and 'No' (Click for fault report.).

SDDJLR218

The screenshot shows a diagnostic interface for the 'ABS PUMP MOTOR POWERFEED TEST'. At the top, there are icons for Vehicle, Settings, and Sessions. Below these are tabs for Session, Selected Symptoms, Related Events, Unrelated Events, Recommendations, and Extras. The main heading is 'ABS PUMP MOTOR POWERFEED TEST'. Underneath, there are links for Pinpoint, Reference, Flow chart, and Options. The test title is 'PWR31-1503 : FAULTY CONNECTOR CHECK.'. The instructions are: 'Switch the ignition off. ABS Module connector disconnected. (FL115)'. The question is 'Are the connectors in good condition?'. The response options are 'Yes' (Go to PWR31-1504) and 'No' (Click for fault report.).

SDDJLR219